

Yi Zheng

Deep Learning Scientist, Thales

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Research and Skills

- Extensive academic and industrial R&D experience in computer vision, machine learning, natural language processing, representation learning, biomedical image processing, biometrics, and multimodal learning.
- State-of-the-art knowledge in statistical graphical models, deep learning (CNNs, Transformers, GNNs), object detection, OCR, semantic multimodal content analysis, vision-language models, and biometric recognition (fingerprint, face).
- Solid skills in prototyping and implementing algorithms and systems on Linux, using Python, C/C++, Matlab, OpenCV, TensorFlow, and PyTorch.

Education

Boston University	Boston, MA
<i>Ph.D. in Computer Science; GPA: 3.93/4.0</i>	2024
University of Southern California	Los Angeles, CA
<i>M.S. in Electrical Engineering; GPA: 3.95/4.0</i>	2016
Shandong Normal University	Jinan, China
<i>B.S. in Electrical Engineering; GPA: 3.90/4.0</i>	2014

Publications

J=Journal, C=Conference, W=Workshop

- [J.1] **Graph attention-based fusion of pathology images and gene expression for prediction of cancer survival** 
Y. Zheng, R. D. Conrad, E. J. Green, E. J. Burks, M. Betke, J. E. Beane, V. B. Kolachalama, *IEEE Transactions on Medical Imaging (IEEE TMI 2024)*
- [J.2] **Graph perceiver network for lung tumor and bronchial premalignant lesion stratification from histopathology** 
R. Gindra, **Y. Zheng**, D. Venkatraman, R. Conrad, E. Green, S. Mazzilli, E. Billatos, M. Reid, E. Burks, V. B. Kolachalama, J. E. Beane, *The American Journal of Pathology (Am. J. Pathol. 2024)*
- [J.3] **A deep learning-based graph-transformer for whole slide image classification** 
Y. Zheng, R. Gindra, M. Betke, J. E. Beane, V. B. Kolachalama, , *IEEE Transactions on Medical Imaging (IEEE TMI 2022)*
- [J.4] **Deep-learning-driven quantification of interstitial fibrosis in digitized kidney biopsies** 
Y. Zheng, R. Gindra, M. Betke, J. E. Beane, V. B. Kolachalama, *The American Journal of Pathology (Am. J. Pathol. 2021)*
- [J.5] **FourierMIL: Fourier filtering-based multiple instance learning for whole slide image analysis** 
Y. Zheng, H. Sharma, M. Betke, J. E. Beane, V. B. Kolachalama, *International Journal of Computer Vision (IJCV 2025 under review)*

- [C.1] **Semantic-Based Sentence Recognition in Images Using Bimodal Deep Learning** 
Y. Zheng, Q. Wang, and M. Betke, *IEEE International Conference on Image Processing (ICIP 2021)*
- [C.2] **LAL: Linguistically aware learning for scene text recognition** 
Y. Zheng, W. Qin, D. Wijaya, and M. Betke, *ACM International Conference on Multimedia (ACM MM 2020)*
- [W.1] **A method for detecting text of arbitrary shapes in natural scenes that improves text spotting** 
Q. Wang, Y. Zheng, and M. Betke, *Computer Vision and Pattern Recognition Workshop (CVPR Workshop 2020)*
- [W.2] **Scraping social media photos posted in Kenya and elsewhere to detect and analyze food types** 
M. Jalal, K. Wang, S. Jefferson, Y. Zheng, E. O. Nsoesie, M. Betke, *Proceedings of the 5th International Workshop on Multimedia Assisted Dietary Management, 2019*

Work Experience

Thales DIS USA, Inc <i>Deep Learning Scientist, Full-time</i> R&D of machine learning and computer vision for: Large-scale biometric systems, and Next-generation contactless fingerprint identification system.	Pasadena, CA 2024-present
Healthcare Co., General Electric (GE) <i>Image Quality Engineer, Full-time</i> Developed the Image-based Collimator Edge Detection (ICED) algorithm which automatically detects collimator edges in x-ray images.	Beijing, China 2016-2017
Brisky, UAV Developer <i>Software Engineer, Intern</i> Enhanced UAV navigation and image stabilization through computer vision-based control algorithms.	Los Angeles, CA Summer 2015

Awards and Honors

Computer Science Research Excellence Award (REA), Boston University, 2022
Masters Honors Fellowship, University of Southern California, 2015
First-class Scholarship Winner, Shandong Normal University, 2011-2014

Professional and Teaching Activities

Reviewer / Program Committee Member

2021	IEEE Access, PETRA, IJIG
2022, 2023	CVPR, ICCV, Nature Communications, Micron
2024, 2025	IEEE Transactions on Medical Imaging, ISCAS, ISBI, CMPB, Journal of Alzheimer's Disease, PBVS Workshop at CVPR

Teaching

BU CS 640: Artificial Intelligence (Graduate Course). Leading TA under Prof. Margrit Betke. Terms: Fall 2017, Fall 2018.
BU CS 132: Linear Algebra (Undergraduate Course). Leading TA under Prof. Abbas Attarwala. Terms: Spring 2018, Spring 2019